

TRAN TAN KHAI

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Education

Ho Chi Minh City University of Science (HCMUS)

Bachelor of Science in Data Science (GPA: 3.47/4.0)

Ho Chi Minh City, VN

Sep. 2024 – 2028

Projects

One Voice Edge | *Python, PyTorch, Edge AI, ONNX, F5-TTS* | GitHub: OneVoice

- Built a 100% offline, real-time bidirectional Speech-to-Speech translation system using multi-threaded **PyTorch** pipelines, thread-safe queues, and **Silero VAD** to minimize idle compute on edge devices.
- Implemented **GIPFormer ONNX (INT8)** via **sherpa-onnx** for industrial noise reduction and Vietnamese **ASR**, and **SenseVoice** for English **ASR** with automatic emotion extraction.
- Integrated **VietAI/envit5-translation** for bilingual translation, utilizing punctuation restoration and custom discourse-marker chunking to minimize latency.
- Deployed Vietnamese **TTS** via **OmniVoice (BetterBox-TTS)** with dynamic emotion mapping, and English **TTS** using **F5-TTS** for voice cloning, optimizing execution to run on **Snapdragon NPU** under 200MB RAM and 600ms latency.

EXACT 2026 Neurosymbolic AI | *LangGraph, Qwen-8B, FastAPI, LlamaIndex* | GitHub: EXACT-2026

- Built a **Neurosymbolic Hybrid AI** agent using **LangGraph** to solve complex college-level Logic and Physics STEM problems, mitigating LLM hallucinations through mathematically verified execution.
- Designed a solver-first reasoning pipeline that dynamically routes questions to **Z3 Theorem Prover** (logic) and **SymPy** (physics), falling back to direct LLM **Chain-of-Thought** reasoning upon solver failure.
- Developed an iterative self-correction loop executing generated solver code in a secure **Subprocess Sandbox**, parsing traceback errors for automatic LLM-driven repair within a strict 60-second latency budget.
- Engineered a semantic retrieval pipeline using **LlamaIndex**, **Qdrant**, and **BAAI cross-encoders**. Deployed via **FastAPI**, optimizing an **8B model** for single **T4 GPU** inference using **BitsAndBytes 4-bit NF4 quantization**.

Financial Text-to-Pandas | *Python, Pandas, BM25, OCR* | GitHub: Financial Text-to-Pandas

- Built an **evidence-first question-answering pipeline** for Vietnamese financial reports, retrieving source documents and OCR tables to answer numerical questions in the **ViFinQA** challenge.
- Implemented **hierarchical BM25 retrieval**, financial value and unit normalization, and schema-aware evidence linking to ground answers in the correct report, table, and cell.
- Designed a typed **CalculationPlan** and constrained **Pandas compiler** with sandboxed execution, producing reproducible answers, executable queries, and source-linked **CSV evidence**.
- Built submission validation around OCR-line provenance and reproducible evidence; the frozen production submission achieved **0.7251 document F2-macro** on the ViFinQA leaderboard.

Technical Skills

Languages: Python, C/C++, SQL (MySQL), R

LLM & Agentic Systems: Qwen, LangChain, LlamaIndex, LangGraph, RAG, Prompt Engineering, Model Quantization (GGUF, NF4), LoRA/QLoRA, Unsloth

Deep Learning: PyTorch, Transformers, CNN, ResNet, U-Net, Mamba, Hugging Face

Machine Learning & AI: Scikit-learn, Reinforcement Learning, Speech Processing (ASR, TTS)

Data Science & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Vector Databases (ChromaDB, Qdrant)

Software Engineering & MLOps: FastAPI, Docker, Git/GitHub, CI/CD, Airflow, MLflow